

EPPAD Bulletin

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Editor's Corner

Dear readers,

EPPAD Bulletin, Volume 6, Number 2 is here. It encompasses new information and scientific articles. The current Fall issue covers various topics which we hope are of interest to readers.

In the News and Highlights section, Dr. Ermias Tilahun summarizes the recent activities of EPPAD. Representatives of EPPAD participated in the 2026 Global Pharmacist Leadership Summit held in Bethesda, Maryland. In this section is also mentioned a work-in-progress: a Memorandum of Understanding between Armauer Hansen Research Institute (AHRI) and EPPAD. The collaboration is envisaged in many areas of common interest, including, but not limited to, traditional medicine initiatives. In this section, the status of the Horn of Africa Clinical Trials (HACT), an EPPAD investment group-incubated group, is reported. HACT has onboarded two local professional partners in Ethiopia to strengthen in-country capacity and execution, is in the process of recruiting officers, completing legal registration and securing a physical facility.

On a different topic, Dr. Asrar Mussa Ahmed is now a Lifetime Member of EPPAD, which is a shining testimony to his commitment to the profession. He is profiled in this edition. The editorial team of EPPAD Bulletin wishes to express appreciation and thanks to Dr. Asrar.

In the Pioneers of Pharmacy section, Ato Aklile gives a detailed account of the history and growth of the Ethiopian Pharmacists Association (EPA). As an insider himself and one of the founding members (as a secretary in 1974) of EPA, he traces back the origin of the association to 1970 when the idea was floated around. It was officially established in 1974. Since its inception, the association has shown tremendous growth not only in membership size (70 in 1974 to the current 5,000), but also in its vision, scope and activities, the article expounds.

Three timely articles are published in this issue. In a “White Paper,” Drs Ermias Tilahun and Elisabeth Gebreegziabher elaborate on real-world data (RWD) and real-world evidence ((RWE) in regulatory science. The paper then goes on to give a detailed discussion of best practices, historical context, selected case studies, and lessons for Ethiopia and other low- and middle-income countries. Pivoting on these concepts, the article argues that Ethiopia can benefit from expansion of these not-as-yet-optimized platforms across health services, and that EPPAD can play a role by partnering with various stakeholders in many areas.

In a review article titled "Ten Medicinal-Chemistry Issues that Restrict Leads from Progressing to Useful Drugs,” Drs. Moa and Tesfaye discuss what they call ten bottlenecks, which relate to the properties of potential drug candidates. The authors provide reasons why and how each of the ten properties blocks development, the typical medicinal-chemistry manifestations, practical chemistry issues, and what all this means in terms of drug development.

In this issue, we also provide Tables (1 and 2) of medicinal plants that were mentioned in an article that was published in Volume 6, Number 1. Although a Table 3 was mentioned in that article, its retrieval was not possible, at least not in due time for this issue. To maintain the integrity of the original paper, the Tables were reproduced as they appeared in the original mimeographed version.

Finally, the Editorial Team would like to encourage readers (professionals both abroad and in Ethiopia) to contribute original, review and other types of articles to EPPAD Bulletin.

Fekadu Fullas, PhD

Editor-in-Chief, *EPPAD Bulletin*

EPPAD News and Highlights

(Prepared by Ermias Tilahun, MPH, PhD; EPPAD President)

EPPAD Represented at the 2026 Global Pharmacist Leadership Summit

EPPAD was proud to be represented at the 2026 Global Pharmacist Leadership Summit, held alongside the National Pharmaceutical Association (NPhA)/Student National Pharmacists Association (SNPhA) Annual Convention in Bethesda, Maryland. Senior Consultants (Board members) Prof. Bisrat Hailemeskel and Ato Akile (Gabriel) Daniel represented EPPAD, joining leaders from diverse and minority pharmacy organizations to foster collaboration, exchange ideas, and build lasting professional partnerships. This participation reflects EPPAD's ongoing commitment to expanding its professional network and amplifying the voice and contributions of Ethiopian pharmacists and pharmaceutical professionals in the diaspora.



EPPAD delegation at the Summit



Engaging with fellow pharmacy leaders

AHRI–EPPAD MOU: Conversation Continues

EPPAD continues active, constructive discussions with the Armauer Hansen Research Institute (AHRI) toward a Memorandum of Understanding to advance shared priorities — including pharmaceutical research and development, traditional medicine standardization, clinical trials, computer-aided drug design, capacity building, and regulatory affairs. The MOU has not yet been signed; the conversation between the two organizations remains ongoing, and EPPAD looks forward to updating members as this partnership progresses toward formalization. This continues EPPAD's track record of formal collaboration in Ethiopia, building on previously signed MOUs with the Ethiopian Ministry of Health (E-MOH), the Ethiopia Food and Drug Administration (EFDA), the Ethiopian Ministry of Education, and the Ethiopian Pharmaceutical Association (EPA).

HACT: Advancing as an Extension of EPPAD's Investment Group

Horn of Africa Clinical Trials (HACT), as an incubated by the EPPAD's investment group, has made steady progress this quarter:

- Onboarded two local professional partners in Ethiopia to strengthen in-country capacity and execution.
- Advanced recruitment for a Chief Executive Officer / Country General Manager to lead operations on the ground.
- Completed legal registration of HACT in Ethiopia.
- Secured an office lease, establishing HACT's physical operational base.

Lifetime EPPAD Member Spotlight: Dr. Asrar Ahmed, PharmD



EPPAD is pleased to welcome Dr. Asrar Mussa Ahmed, PharmD, as a Lifetime Member. Originally from Addis Ababa, Dr. Ahmed earned an Associate Degree in Pharmaceutical Science

(2002) and a Bachelor's Degree in Pharmacy (2010) from Jimma University before relocating to the United States. He later completed pre-pharmacy studies at Georgia State University and earned his Doctor of Pharmacy (PharmD) from South University in 2025. His clinical pharmacy experience spans internal medicine, neurology, psychiatry, endocrinology, HIV, and other infectious diseases, with a strong pharmacotherapy focus. Dr. Ahmed currently practices as a full-time community pharmacist in Georgia, where his responsibilities include comprehensive pharmaceutical care, medication therapy management, drug utilization review, immunizations, and supervision of pharmacy staff. EPPAD is grateful for his commitment as a Lifetime Member and looks forward to his continued engagement with our professional community.

Thank you for your continued support of EPPAD's mission.

Dr. Ermias Tilahun, PhD, MPH

President, EPPAD

Pioneers of Ethiopian Pharmacy and Related Fields

Aklile G Giorgis (aka Gabriel Daniel)

In this issue of EPPAD Bulletin we have selected to feature the Ethiopian Pharmaceutical Association (EPA), one of the oldest professional health associations in Ethiopia. Its development reflects the broader evolution of pharmacy practice and health systems in the country.



Key Summary Historical Milestones

- **1970–1974: Formation & Official Establishment:** Initiated by pharmacy students and a group of 40 pioneering pharmacists, the EPA officially registered in November 1974 as the representative body for Ethiopian pharmacists.
- **1978–1979: Internal Reorganization:** Reorganized its institutional framework to align pharmacy services with national public health priorities during the socialist transition period.
- **1991: Securing Professional Autonomy:** Re-established as an autonomous, self-governing non-governmental organization under Ministry of Justice Registration No. 177, shedding direct government control.
- **Launch of Academic & Professional Standards:** Created the *Ethiopian Pharmaceutical Journal (EPJ)* to publish indigenous pharmaceutical research, alongside authoring the national Code of Ethics for Pharmacists.
- **Curriculum Modernization Advocacy:** Led the push alongside university faculties to transition the national university curriculum from a traditional drug-dispensing model to a clinical, patient-centered pharmacy practice model.
- **Mandatory CPD Platform Integration:** Partnered with the World Continuing Education Alliance (WCEA) and the Ministry of Health to roll out a nationwide, app-based Continuing Professional Development (CPD) system required for annual license renewals.
- **2024: Golden Jubilee (50th Anniversary):** Celebrated 50 years of continuous operation (1974–2024), honoring international partners like the WHO for contributions to regulatory development, local production, and combating Antimicrobial Resistance (AMR).
- **2025: Launch of Multipurpose Headquarters Project:** The multistory multipurpose complex will provide a permanent institutional hub for research, conference hosting, and financial sustainability.

The history of the Ethiopian Pharmaceutical Association (EPA) begins in 1970, a small group of visionary Ethiopian pharmacists initiated the concept of a national professional association. That same year, pharmacy students at Haile Selassie I University (now Addis Ababa University) formed the *Students Pharmaceutical Association*, creating an organizational blueprint for the professional body.

The Ethiopian Pharmaceutical Association (EPA) “የኢትዮጵያ የመድኃኒት ቅመማ ማህበር *Ye Etyopiya Yemedehanit Kemema Mahaber*” (in Amharic) was officially established and legally registered in November 1974 by a founding group of 40 pharmacists. This period coincided with the historic political unrest following the Ethiopian Revolution and the collapse of the imperial regime, making its operating years extremely challenging.

The highest governing body of this maiden association was the General Assembly with an Executive Committee consisting of seven members running the day-to-day affair and serving one year term.

The first Executive Committee of EPA in 1974 consisted of the following seven elected officers:

- Belachew Desta (National University) – President
- Birhanu Zeleke (EPHARM) - V/ President
- Akile G. Giorgis (Menilik II Hospital) – Secretary
- Belaynesh Desta (Wavel Pharmacy) – Treasurer
- Girma Daffa (Jos Hansen) - Public Relation Officer
- Berhe Seyoum (Jos Hansen) – Member
- Baro Tumsa (Inquiry Commission) - Member

Membership of the association consists of different categories: Regular, Honorary, Associate, Student, Traditional Medicine, Institutional.

Although navigating through major political shifts over the last five decades required the EPA to continuously adapt its institutional posture, legal status, and strategic focus, it has achieved key milestones that have fundamentally transformed pharmacy practice, regulatory standards, and health policy across Ethiopia.

The major challenge faced by EPA was following Emperor Haile Selassie’s overthrow by the Derg in the 1974 revolution. During the first decade of the socialist regime all professional associations were relegated to fall under an umbrella Federation of Health Professional Association and EPA became a semi-independent pharmacy entity within the Federation losing its autonomy. Yet again, in 1984 the limited semi-independence enjoyed by the individual health professional associations came under serious restriction when all the associations were finally forced to fall under the consolidated constitution as one entity under the Union of the Ethiopian Health Profession Union.

With the overthrow of the Derg by EPDRF in May 1991, EPA officially re-established itself as an autonomous, self-governing entity. It secured legal registration (No. 177) from the Ministry of Justice, breaking away from historical state oversight. The association expanded its mandate beyond basic state support to actively shape modern pharmaceutical policy, authoring professional codes of ethics, and working alongside regulatory bodies like DACA/EFMHACA (now EFDA).

Vision (*Strategic advocacy, long-term policy shaping, and global partnerships*)

- **Policy Reform & Advocacy:** Advocates for pharmaceutical development and contributes directly to national drug policy formulations and sectorial concept papers.
- **Global & Diaspora Partnerships:** Partners with international groups—such as the Ethiopian Pharmacists and Pharmaceutical Scientists in Diaspora (EPPAD)—to advance knowledge exchange, drug discovery, and traditional medicinal plant research.
- **Inter-Agency Collaboration:** Maintains long-term partnerships through committees and working groups across government offices, health facilities, and industry partners.

Mission (*Core operational functions and immediate sectoral contributions*)

- **Education & Professional Development:** Collaborates with university faculties to structure ward attachments, community pharmacy clerkships, and industrial placements for final-year students. Designs accredited e-learning and hands-on Continuing Professional Development (CPD) modules to keep practicing pharmacists updated.
- **Knowledge Sharing & Academic Publishing:** Publishes the *Ethiopian Pharmaceutical Journal (EPJ)*, *Pharma Forum*, and *Book of Abstracts*. Hosts Ethiopia's largest annual scientific gathering of researchers, clinical specialists, manufacturers, and regulators.
 - **Regulatory & Standard Operating Procedures (SOPs) & Practice Manuals:** Develops and enforces the code of ethics for the pharmacy profession. Serves on technical working groups with the Ethiopian Food and Drug Authority (EFDA) and the Ministry of Health to regulate Good Dispensing Practices (GDP) and storage guidelines and update essential medicine lists; Hospital and community pharmacy practice models.
- **Public Health Initiatives:** Leads national campaigns against Antimicrobial Resistance (AMR) and conducts public screening initiatives for non-communicable diseases like hypertension and diabetes.
- **Transition to Clinical Pharmacy:** EPA partners closely with major institutions—such as the Addis Ababa University (AAU) School of Pharmacy—to modernize academic curricula. The association played a key role in shifting the national pharmacy curriculum from a purely product-focused model toward patient-centered clinical pharmacy.
- **Workforce Needs Alignment:** EPA provides regular feedback from practicing pharmacists to universities, ensuring undergraduate and postgraduate programs adapt to national

priority areas like oncology, rational drug use, and supply chain logistics.

- **Code of Ethics for Ethiopian Pharmacists:** Guidance defining legal responsibilities, professional boundaries, and patient-centered ethics for practicing pharmacists. Delivers mandatory license-renewal credits via digital courses and workshops.

Membership:

The category of membership of EPA is made up of:

- **Regular Members** (individuals who have graduated with a Bachelor or above in pharmacy from a recognized university or similar institutions).
- **Associate Members** (individuals with diploma in pharmacy from a recognized university or similar institutions, those registered as Druggists by the Ministry of Health, last year students of schools of pharmacy, individuals with at least a bachelor's degree in different branches of science and recognized by the General Assembly to have an outstanding contribution to the development of the pharmacy profession).
- **Traditional Medicine Practitioners** (with recognized contribution to the development of pharmacy profession)
- **Institution Members** (Organizations/Institutions working in health areas and pharmaceutical sector and recognized by the General Assembly to contribute to the development of the profession).
- **Honorary Members** (individuals who have made outstanding contributions in traditional medicine, profession of pharmacy and other outstanding contributions)

During its establishment in 1974, the founding and registered members of the association were 70. Membership records from the period after its re-registration in 1993 were 130. And only 130 until 1993. In 2019, membership reached 2,736. Current membership climbed up to 5,000.

Management and Office Base

The highest body of EPA as a professional association is its Board of Directors which has a Central Executive Body. The Executive Director who reports to the Central Executive Body heads an office

of about half a dozen staff who manage the day-to-day business of the association.

Current EPA branches are established in Awassa, Adama, Dire Dawa, Jimma, Bahrdar, Gondar, Dessie, and Mekele.



Picture showing the current office and planned 10 story multipurpose headquarters to be built at the same location

The Association is currently using an old residential building it bought several years ago as its office

The **EPA Multipurpose Headquarters Project** (2B+G+9 building) represents a major infrastructure milestone for the Ethiopian Pharmaceutical

Association (EPA). It provides the EPA with an iconic physical presence, symbolizing the professional maturation of pharmacy practice in Ethiopia. Aimed at establishing a centralized physical, operational, and research hub in Addis Ababa, the project transitions the association from rented or administrative office spaces into an autonomous, state-of-the-art facility. It incorporates rentable office floors, commercial spaces, and event venues to generate recurring revenue for EPA programs. It is planned to house Houses dedicated training halls, digital learning studios, and conference rooms to host accredited workshops, e-learning production, and skill updates for practicing pharmacists. It will provide space for a physical and digital pharmaceutical library, archives, and research databases to serve university students, researchers, and policymakers. Revenue generated from leasable spaces directly funds student scholarships, research grants, public health campaigns (e.g., Antimicrobial Resistance awareness), and regional chapter developments.



EPA has formed a Board of Ambassadors to plan and implement the project. It is reaching out to family and friends of EPA to donate to this historic project.

EPA has reached out to EPPAD members for opportunities to contribute through advocacy, partnership, and resource mobilization.

Conferences

Conferences and symposia organized by the **Ethiopian Pharmaceutical Association (EPA)** serve as the central platform for pharmaceutical scientists, clinical pharmacists, regulatory experts, and industry stakeholders. It holds regular annual

scientific conference and recently held its 50th year Golden Jubilee Celebration.

(Photo showing HIV/AIDS programming multisector panelists)

Partnerships

In its five decades of existence, EPA has partnered with **Ministry of Health (MoH) & Regional Health Bureaus** on national pharmaceutical supply chains, essential medicine lists, and public health campaigns from its inception **Ethiopian Food and Drug Authority (EFDA)** on shaping pharmaceutical regulations, enforcing Good Dispensing Practices (GDP), monitoring drug quality, and implementing continuing professional development (CPD) requirements. **Ethiopian Pharmaceuticals Supply Service (EPSS)**: Works jointly on logistics, warehousing standards, and distribution policies for essential medicines nationwide **International Pharmaceutical Federation (FIP)**: Engages with global pharmacy leaders to align Ethiopian clinical practices with international workforce and practice standards.

East African Pharmacists Federation (EAPF):

Collaborates with regional partners across East Africa to harmonize pharmacy education, cross-border regulation, and practice guidelines.

World Continuing Education Alliance (WCEA):

Jointly provides an online platform and mobile app offering free, accredited CPD courses and digital certification for Ethiopian pharmacists. **Universities (e.g., Addis Ababa University School of Pharmacy)**: Partners on curriculum design, hosting scientific conferences, and supporting academic research published in the *Ethiopian Pharmaceutical Journal*.

Local Pharmaceutical Manufacturers (e.g., EPHARM):

Collaborates on joint research initiatives—such as drug stability studies and formulation developments—and advocates for national manufacturing capacity.

Ethiopian Pharmacists and Pharmaceutical Scientists in Diaspora (EPPAD):

Works together on community outreach, philanthropy, student scholarships, equipment donations, and scientific knowledge exchanges.

Global Health NGOs (e.g., USAID, CHAI, WHO):

Partners on grant-funded projects focusing on antimicrobial resistance (AMR), rational drug use, tuberculosis/HIV pharmaceutical care, and capacity building.

EPA is a Member in Different Agencies and Tasks

- National Health Professionals Ethics Committee,
- Health Professional Associations Consortium at the Ethics and Anticorruption Commission,
- Task Force Working for the Establishment of Health Professional Council in Ethiopia,
- National Antimicrobial Resistance Advisory Committee,
- Health Facilities Standard Development Committee at the Standards Agency,
- Examination Development and Oversight Committee at the FMOH,
- International Pharmaceutical Federation (FIP).

Core Challenges Faced by the EPA Since Inception

- Low members' commitment,
- Limited managerial, technical, material and physical capacity,
- Limited financial resources,
- Inadequate involvement in maintaining the quality of pharmaceutical education, research and practice,
- Unsatisfactory cooperation with stakeholders, collaborators and the public,
- Inadequate involvement or participation in the development of policies, laws, and directives related to pharmacy profession.

Core Academic & Scientific Publications

The Ethiopian Pharmaceutical Association (EPA) produces peer-reviewed scientific literature, ongoing practice guides, and operational standards for practitioners.

- **Ethiopian Pharmaceutical Journal (EPJ) (ISSN: 1029-5933):** The flagship academic journal associated with the national pharmaceutical community. It publishes original peer-reviewed research across all disciplines of pharmaceutical science,
- **Annual Scientific Conference Books of Abstracts:** Published in conjunction with the EPA's regular national conferences, documenting emerging clinical findings, health economics data, and policy research presented by Ethiopian researchers.
- **Pharma Forum / Professional Newsletters:** Newsletters and trade publications distributed to members covering regional market updates, association activities, regulatory updates, and workforce news.

Future Planned Activities & Strategic Focus Areas

The EPA's current multi-year strategic vision focuses on digital transformation, strengthening local manufacturing, and expanding professional advocacy:

- **Advancing Digital Health & Tele-Pharmacy:** Expanding e-learning platforms for Continuing Professional Development (CPD) in partnership with international bodies (such as WCEA) and integrating AI/digital tools into supply chain oversight and patient counseling.
- **Supporting Local Pharmaceutical Manufacturing:** Partnering with the government to transition Ethiopia from an import-dependent market to a regional production hub, specifically supporting initiatives at dedicated industrial parks (such as the *Kilinto* Special Economic Zone).
- **Antimicrobial Stewardship (AMS) & Public Health Programs:** Scaling national campaigns and training modules focused on combating Antimicrobial Resistance (AMR), rational drug use, and non-communicable disease (NCD) management.

- **Strengthening Regulatory Compliance & Quality Control:** Working in tandem with the Ethiopian Food and Drug Authority (EFDA) to enforce stricter post-marketing surveillance, combat counterfeit/substandard medications, and raise standards to achieve higher WHO regulatory maturity levels.
- **Expanding Regional & Global Footprints:** Deepening ties with the East African Pharmacists Federation (EAPF) and the International Pharmaceutical Federation (FIP) to harmonize regional drug standards and facilitate international knowledge transfer

The Ethiopian Pharmaceutical Association since 2014 has been observing Pharmacy Day program to complement the annual conference which more dwells on official programs and scientific issues while the pharmacy day will focus on discussions of pressing and current professional issues. Such program of celebrating Pharmacy Day is also observed by the International Pharmaceutical Federation (FIP) on observing the day on November 17th (*Hidar* 25th EC of the Ethiopian calendar), the day of EPA's establishment.

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Real-World Evidence for Regulatory Submissions and Public Health Decision Best Practices, Historical Context, Selected Case Studies, and Lessons for Ethiopia and Other Low- and Middle-Income Countries

Ermias Tilahun, MPH, PhD, and Elisabeth Gebreegzabher, PhD

Purpose of this white paper

This white paper takes stock of where real-world data (RWD) and real-world evidence (RWE) stand today in regulatory science. It traces how the field arrived here, draws out best practices from recent guidance and real submissions, and asks a more grounded question: what does this actually mean for Ethiopia and other low- and middle-income countries (LMICs)?

The message underneath all of it is simple. RWE can genuinely strengthen regulatory, clinical, coverage, and public health decisions, but only when it is planned early, built on data that fits the question at hand, governed with care, and analyzed using methods that are transparent and locked in before anyone looks at the results.

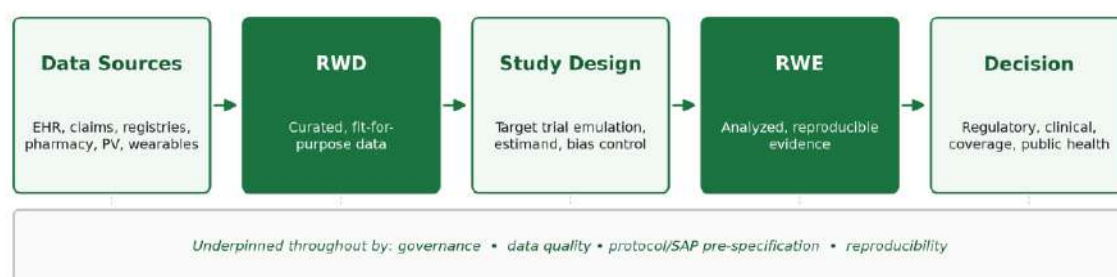


Figure 1. Conceptual framework: real-world data becomes regulatory-grade real-world evidence through deliberate study design and bias control, governed throughout by quality and reproducibility standards, to inform regulatory, clinical, coverage, and public health decisions.

1. Executive Overview and Historical Background

Real-world evidence has outgrown its old reputation as a nice-to-have on the margins of clinical development. When it is tied to a clear decision and built on data that is relevant, reliable, transparent, and reproducible, it now carries real weight in regulatory science. Real-world data (RWD), information gathered outside the controlled setting of randomized trials, comes from electronic health records, insurance claims, registries, pharmacy records, laboratory systems, biobanks, wearable devices, and pharmacovigilance databases. RWE, in turn, is simply the clinical evidence that emerges once that data is analyzed.

The field's roots are in safety surveillance after a drug reaches the market, and that is still where most people's mental picture of RWD begins. But its applications have since expanded across the entire product lifecycle, from informing clinical trial planning to supporting external control arms for single-arm trials, natural history studies, effectiveness research, label expansion, health technology assessment, and coverage decisions. The clearest

turning point came with the 21st Century Cures Act, which pushed RWE onto the FDA's formal agenda and led to a framework and a string of guidance documents covering data sources, study design, data standards, registries, EHR and claims data, externally controlled trials, and non-interventional studies.

That growth has cut both ways. It has opened doors, but it has also raised the bar for what counts as credible evidence. A large dataset, or one that is simply convenient to access, is not enough on its own. Regulators want to know whether the data can actually answer the question being asked, whether the key variables hold up, whether bias has been dealt with honestly, and whether another reviewer could retrace the analysis and reach the same conclusions.

For Ethiopia and other LMICs, that is good news. Building a perfect national electronic health system is not a prerequisite for putting RWE to work, countries can start smaller and still make real progress. Priority questions, sentinel hospitals, disease registries, pharmacy and laboratory data streams, pharmacovigilance systems, and well-governed pilot projects focused on high-burden diseases, medicine safety, vaccine rollout, and access decisions are all realistic starting points. The table below maps out where this kind of evidence tends to carry the most weight.

RWE use area	How it supports decision-making
External control arms	Gives single-arm trials a point of comparison when randomizing patients is not feasible or ethical.
Natural history and burden of disease	Maps out how a disease typically progresses, where the unmet needs lie, baseline risk, and which endpoints actually matter.
Post-approval safety	Catches rare or delayed adverse events once a medicine is used more broadly in routine care.
Label expansion and new indications	Backs the case for use in new populations, provided the evidence is credible and the methods were locked in beforehand.
Coverage and HTA	Helps payers and health systems understand effectiveness, safety, utilization, and value as they play out in real practice.
Clinical trial planning and feasibility	Helps sponsors estimate eligible patient populations, understand current treatment patterns, refine study protocols, select sites, and plan recruitment.

2. Regulatory Landscape and Shared Expectations

Food and Drug Administration (FDA), European Medicines Agency (EMA), (International Council for Harmonization of Technical Requirements for Pharmaceutical for Human Use (ICH), and Medicines and Healthcare products Regulatory Agency (MHRA) do not always speak with one voice, but on RWE they are converging on the same core principles: engage regulators early, use data that is fit for purpose, report transparently, lock in protocols and statistical analysis plans before analysis begins, document where the data came from, and address bias with a clear method rather than an afterthought. In practice, that means sponsors and researchers cannot treat RWE as a shortcut. It deserves the same rigor as any clinical development program, even when the underlying data was collected during routine care rather than a trial.

This is where a solid best-practice framework earns its keep: it maps requirements across the entire study lifecycle — design, protocol and Statistical Analysis Plan (SAP) development, data evaluation, curation and transformation, analysis, reporting, and delivery. That kind of structure matters especially in LMIC settings, where credibility rests on more than statistical technique alone. Operational discipline, documentation, data management, quality control, and a regulator's ability to verify the evidence all carry real weight.

Externally controlled trials sit at the top of the evidentiary ladder, and for good reason: they ask most of the data. They work best when a disease's natural history is well understood, when the external control group closely resembles the treated patients, when concomitant treatments do not differ in meaningful ways, and when the treatment effect is large enough to clearly stand apart from how the disease would have progressed anyway. That combination shows up most often in rare diseases, fast-moving illnesses, serious infections, oncology, and situations where withholding treatment to run a traditional trial would be unethical. The table below shows what these shared expectations look like once they are put into practice.

Regulatory expectation	Practical meaning for RWE studies
Early engagement	Talk to regulators before the study starts: rationale, intended use, data sources, design, protocol, SAP, and what the final data delivery should look like.
Data quality and fitness	Define the research question first, then choose the data, and be ready to explain why that source can capture the population, exposure, comparator, outcome, follow-up, and covariates you need.
Design robustness	Lean on target trial emulation, active comparators where feasible, eligibility criteria that mirror a real trial, appropriate endpoints, and methods that genuinely address confounding and selection bias.

Regulatory expectation	Practical meaning for RWE studies
Data management	Keep source data traceable, maintain audit trails and data dictionaries, validate your variables, use 21 Code of Federal Regulations (CFR) Part 11-compliant systems where they apply, and follow standards such as Clinical Data Interchange Standards Consortium (CDISC) for anything headed to regulators.
Reproducibility	Lock in the protocol and SAP before touching the data, and submit patient-level data and analysis datasets when asked, so regulators can rerun the key analyses themselves.

3. Practical Methods for Regulatory-Grade RWE

Credible RWE starts with a question the available data can answer, not the other way around. The single most important discipline here is defining the target trial before anything else: the eligibility criteria, treatment strategies, index date, comparator, outcome, follow-up period, and analytic approach all need to be settled before anyone looks at results. Skip this step, and the door opens to post hoc decision-making and to a study regulators struggle to follow.

It also helps to define the estimand up front, using the five ICH E9(R1) attributes: population, treatment, endpoint, intercurrent events, and summary measures. Doing so keeps the clinical question, the RWD emulation, and the regulatory interpretation pointed in the same direction. A protocol and SAP locked in before analysis begins remains the best defense against selective reporting and against hypothesizing after the results are known (HARKing).

Bias control is not one-size-fits-all; it has to match whatever threat is present in the data. Confounding might call for propensity score matching or weighting, doubly robust estimators, or negative control outcomes. Selection bias often calls for new-user active-comparator designs, explicit eligibility criteria, and real vigilance against immortal time bias. Information bias demands validated definitions for exposure and outcome, both positive and negative controls, and a systematic look at what is missing from the data. The table below breaks these methods down by the specific risk each one addresses.

Method area	Regulatory-grade expectation
Feasibility before data selection	Weigh every plausible data source, document why each one was accepted or ruled out, and justify whichever source you land on.

Method area	Regulatory-grade expectation
Variable validation	Validate key exposure, outcome, and covariate definitions ahead of time whenever those variables are new or not already validated in routine use.
Comparator strategy	Make sure comparator groups are clinically and temporally alike before modeling even begins; statistical adjustment alone is not a substitute.
Sensitivity and uncertainty	Stress-test your assumptions with sensitivity analyses, negative controls, quantitative bias analysis, and a real look at missing data.
Inspection readiness	Keep source records, researcher logs, documentation of every data transformation, QC records, and audit trails ready for inspection.

4. Selected Global Regulatory Case Studies

A handful of recent FDA case studies capture both what RWE can do well and where it can go wrong. The lesson that comes through repeatedly is that successful submissions are not improvised in the final stretch of development. They are planned from the start, discussed openly with regulators, and backed by data and methods clear enough for reviewers to understand and, if needed, reproduce. Four cases below show what that looks like in practice, for better and for worse.

Case	RWE role and key lesson
Lumakras® (sotorasib)	RWE served as supporting evidence, mapping the natural history and outcomes of Kirsten rat sarcoma viral oncogene homolog (KRAS) G12C-mutated advanced Non-Small Cell Lung Cancer (NSCLC). What made it work was drawing on multiple fit-for-purpose data sources and seeing consistent results across cohorts.
Viojoice® (alpelisib)	Here, RWE carried real weight as primary evidence for a rare disease indication, built on a retrospective single-arm study drawn from a compassionate use program. The best practices that made it credible: early FDA engagement, a protocol and SAP defined ahead of time, independent assessment of endpoints, conversion to CDISC standards, patient-level data, and readiness for inspection.

Case	RWE role and key lesson
Prograf® (tacrolimus)	Registry-based RWE, together with published literature, supported a label expansion for lung transplant, an area where running a randomized trial would no longer be ethical. It is a good illustration of what national registry data can do when paired with a predefined protocol and SAP, patient-level data, and confirmatory evidence from indications already approved.
Omblastys® (omburtamab)	This one serves as a cautionary tale: concerns about comparator suitability, endpoint choice, immortal time bias, and comparability between arms were never fully resolved. Early conversations with regulators matter, but they are not enough on their own; their advice actually has to be acted on.

5. Selected RWD/RWE Case Studies in Africa

Africa has its own set of examples worth studying closely, where real-world data shaped decisions on safety, effectiveness, implementation, policy, or coverage. Not all of these were formal marketing authorization filings, but that is almost beside the point. What makes them valuable is what they show together: that regulatory and public health evidence systems can be built from routine data, registries, surveillance, and implementation platforms to support major decisions — as long as those systems are designed and governed well. The examples below span several countries, health areas, and evidence-generation approaches.

Africa case study	RWD/RWE contribution and best-practice lesson
Botswana Tsepamo surveillance and dolutegravir safety	Birth-outcome surveillance picked up a possible neural tube defect signal in women exposed to dolutegravir around the time of conception. As more evidence came in and the benefit-risk picture became clearer, World Health Organization (WHO) guidance ultimately recommended dolutegravir as the preferred first- and second-line HIV treatment for everyone, including pregnant women and women who could become pregnant. Lesson: prospective safety surveillance with a known denominator can shape global treatment policy quickly, provided the risk-benefit interpretation stays transparent throughout.

Africa case study	RWD/RWE contribution and best-practice lesson
repeat T epitopes S antigen (RTS,S) malaria vaccine pilots in Ghana, Kenya, and Malawi	WHO's 2021 recommendation to roll out RTS,S widely was based on pilot data gathered from routine child health clinics in Ghana, Kenya, and Malawi. Those pilots showed the vaccine was feasible to deliver, reached children equitably, was safe, did not undercut other preventive behaviors, and cut severe malaria by 30% in real-life conditions. Lesson: implementation-focused RWE can close the gap between what a trial shows and what a country is willing to adopt as policy.
MenAfriVac impact evaluation in Chad and the meningitis belt	After mass vaccination in Chad, post-introduction surveillance and a community evaluation showed a sharp drop in serogroup A meningitis and in carriage of the bacteria. Lesson: combining disease surveillance with carriage and incidence monitoring can demonstrate real population impact, justify continued investment in immunization, and flag early warning signs of serogroup replacement.
South Africa COVID-19 data and vaccine implementation experience	During COVID-19, South Africa leaned on quickly curated public data, dashboards, and implementation studies to guide both public health decisions and vaccine rollout. Lesson: even imperfect, real-time data can support good policy, as long as it is transparent, kept current, and actually tied to operational decisions on the ground.
Burkina Faso passive clinic surveillance embedded within a cluster-randomized trial	As part of the Community Health with Azithromycin Trial (CHAT), routine diagnosis and treatment data were systematically captured from sentinel health facilities to assess secondary endpoints and answer additional research questions beyond the trial's original objectives, ultimately generating multiple peer-reviewed studies (Sié et al., CHAT protocol, 2019; Oldenburg et al., 2021). Lesson: research-grade RWD can be built incrementally from well-governed sentinel sites without requiring a national electronic health record system, creating reusable data assets for future clinical and public health research.

6. Practical Ethiopia/LMIC Use Case: Building a Disease-Specific RWE Platform

Ethiopia illustrates both the promise and the challenge of building regulatory-grade RWE in an LMIC setting. The country has invested substantially in digital health, including platforms such as SmartCare and OpenMRS, and continues working toward a more interoperable electronic medical record ecosystem. Even so, routine clinical information remains fragmented across hospitals, laboratories, pharmacies, and hybrid paper-electronic systems (Ethiopian MoH, Digital Health Portal; Tilahun & Fritz). As a result, observational researchers often rely on

disease registries or aggregate statistics that cannot fully characterize treatment trajectories, standards of care, or longitudinal outcomes.

The Addis Ababa City Cancer Registry (AACCR) illustrates both this gap and the opportunity. A study of AACCR-registered patients found that treatment information was missing from the registry and had to be retrospectively abstracted from individual hospital records to complete the therapy picture (Feuchtner et al., 2019). This reflects a broader challenge: clinically valuable patient-level information often already exists within routine healthcare systems but has not yet been standardized, linked, or organized into research-ready datasets. The opportunity for Ethiopia and other LMICs is therefore not necessarily to generate entirely new sources of data, but to develop the methods, governance, and quality systems needed to transform existing clinical information into reproducible, regulatory-grade evidence.

Building on this foundation, what might a practical approach look like for Ethiopia specifically? Picture a disease-specific RWE platform built around high-burden or neglected conditions: visceral leishmaniasis, schistosomiasis, antimicrobial-resistant infections, vaccine-preventable diseases, or maternal and child health. It would not need to start as a sweeping national electronic health record system: a handful of referral hospitals, laboratories, pharmacies, treatment centers, and pharmacovigilance reporting points would be enough to get moving. Depending on the clinical question and the available infrastructure, this could begin either through structured abstraction of existing clinical records or through prospective data collection embedded within routine care.

Say Ethiopia rolls out a new treatment regimen for visceral leishmaniasis or schistosomiasis. A prospective registry could define, before analysis ever begins, the eligibility criteria, index date, treatment exposure, comparator group, outcomes, follow-up schedule, adverse event reporting, laboratory confirmation, and data quality checks. Linking that registry to pharmacy dispensing data, laboratory results, and safety reports could then support effectiveness assessment, relapse monitoring, subgroup analysis, post-approval safety work, and planning for future trials.

The same approach adapts easily to vaccines, oncology access programs, chronic disease medicines, pediatric medicines, and maternal health interventions. For the Ethiopian Pharmacists and Pharmaceutical Scientists Association in the Diaspora (EPPAD), the near-term opportunity is less about technology and more about building people and processes. That means helping develop the templates and governance that make this kind of data usable: registry protocols, data dictionaries, variable definitions, monitoring plans, pharmacovigilance workflows, statistical analysis plans, and reporting templates built for reproducibility. The table below breaks this down into concrete building blocks.

Building block	Minimum practical action
Priority question	Start from an actual decision that needs to be made; regulatory approval, safety monitoring, guideline update, reimbursement, access, or trial planning.
Fit-for-purpose data source	Choose sentinel sites or disease registries where diagnosis, treatment, laboratory confirmation, outcomes, and follow-up can all actually be measured.

Building block	Minimum practical action
Data capture and standardization	Develop standardized variable definitions, case report or standardized abstraction forms, coding practices, and documentation standards so key exposures, outcomes, laboratory results, and follow-up are captured consistently across sites and can be linked and reused.
Data quality governance	Build data dictionaries, validate key variables, implement QC checks, conduct a missingness review, and maintain source documentation, audit trails, and data provenance before analysis starts, not after, to ensure traceability and reproducibility.
Bias control	Identify the main sources of bias and apply appropriate methods, including target trial emulation, active comparators when possible, eligibility criteria set in advance, sensitivity analyses, and negative controls.
Capacity development	Invest in people: train teams in pharmacoepidemiology, biostatistics, data management, regulatory science, pharmacovigilance, and quality systems.

7. Lessons for Developing Countries and EPPAD Action Agenda

If there is one lesson developing countries should take from all of this, it is that RWE capacity needs to be built around decisions, not technology for its own sake. A well-governed registry running at three high-quality sentinel hospitals will often do better than a sprawling database nobody has properly documented. Regulatory-grade evidence has less to do with volume and everything to do with clarity, discipline, and the ability to trace every step back to its source.

Ethiopia and countries like it are better served starting small: a handful of high-value pilots where the evidence gap is obvious and local data can realistically be collected to a reliable standard. The strongest candidates tend to be areas with a heavy disease burden, thin trial data, pressing safety questions, or major implications for access and coverage. Given time, these pilots can grow into something closer to a national or regional RWE network.

One practical pathway for LMICs is to begin with targeted, well-governed data collection rather than waiting for comprehensive national electronic health record systems. For many regulatory, clinical, and public health questions, researchers do not need immediate access to millions of electronic records. Carefully designed retrospective chart abstraction can transform existing clinical documentation into fit-for-purpose datasets, while prospective data collection at selected sentinel sites can capture key variables systematically when they are not routinely available. Both approaches provide practical entry points for generating evidence on disease burden, treatment patterns, standards of care, outcomes, medicine safety, and healthcare utilization.

The appropriate approach depends on the clinical question and the maturity of the available data environment. Retrospective abstraction can rapidly unlock value from existing healthcare records, whereas prospective approaches strengthen consistency, follow-up, and data completeness over time. Together, the two provide a pragmatic bridge between routine clinical care and regulatory-grade evidence generation, supporting applications that range from natural history studies to post-approval safety monitoring, several of which are discussed below.

EPPAD has a role to play at several of these stages, summarized below.

EPPAD contribution	Potential activity
Training	Run short courses covering RWE, causal inference, registry design, pharmacovigilance, data quality, and reproducible analysis.
Templates	Build practical, ready-to-use templates for registry protocols, SAPs, data dictionaries, feasibility assessments, and RWE study reports.
Partnerships	Bridge diaspora expertise with Ethiopian Food and Drug Administration (EFDA), universities, hospitals, research centers, professional societies, and patient groups.
Pilot projects	Back disease-focused RWE pilots across infectious disease, vaccines, oncology access, chronic disease medicines, and maternal-child health.
Quality culture	Champion a culture built on pre-specification, audit trails, transparent reporting, validated outcomes, and early engagement with regulators and stakeholders.

For countries or organizations seeking to build sustained RWE capability, development may follow a staged progression:

Phase	Practical objective
Phase 1	Establish standardized chart-abstraction capability at a small number of sentinel hospitals to answer specific clinical and regulatory questions through commissioned projects or pilot studies.
Phase 2	Expand into recurring RWE activities, including disease registries, longitudinal cohorts, natural history studies, treatment-pattern analyses, comparative-effectiveness research, and post-market safety monitoring.

Phase	Practical objective
Phase 3	Introduce technology-enabled approaches including interoperable data systems, AI-assisted abstraction, and natural language processing to improve efficiency once standardized workflows and quality systems are established.

This staged progression reflects one common pathway through which RWE organizations have developed internationally, often starting with expert human abstraction of clinical records to create research-quality datasets before incorporating machine learning and natural language processing to improve efficiency and scale (IPM, Flatiron Health EHR Data Extraction). The broader lesson is that technology amplifies well-designed processes but rarely substitutes for them. Strong governance, standardized abstraction, and rigorous quality control remain the foundation of regulatory-grade evidence.

AI-assisted abstraction deserves particular attention given how quickly it is advancing. Natural language processing and large language models can now pull structured variables — diagnoses, treatment dates, lab values, staging — directly from unstructured clinical notes, cutting abstraction time and cost compared with fully manual review. The caveat is that these tools still require human-in-the-loop validation: a sampled comparison against source documents, documented error rates, and a clear audit trail distinguishing machine-derived from human-confirmed fields. For Ethiopia, this points to a realistic near-term role for AI as an accelerant within Phase 1 and Phase 2 sentinel-site abstraction, not a replacement for the governance and quality-control foundations described above.

8. Challenges and Limitations

Even well-designed RWE efforts face real constraints and naming them plainly is part of the discipline. Four recurring challenges shape what is realistic for Ethiopia and other LMICs in the near term:

Challenge	Practical implication
Data quality	Incomplete, inconsistent, or unvalidated records limit which questions RWD can credibly answer; abstraction and validation take real time and expertise.
Interoperability	Hospital, laboratory, pharmacy, and pharmacovigilance systems rarely share common identifiers or coding standards, which complicates linkage and follow-up.
Governance	Data-sharing agreements, privacy protections, and ethical oversight need to be established before data collection begins, not retrofitted afterward.

Challenge	Practical implication
Workforce capacity	Few local specialists are trained in pharmacoepidemiology, biostatistics, and regulatory science; building this pipeline takes sustained investment.

None of these constraints are unique to LMICs, but they carry more weight where infrastructure and specialized training are still developing.

9. Future Directions

Looking ahead, a few developments are likely to shape RWE's trajectory in Ethiopia and comparable settings. Ongoing interoperability efforts, including EMR and health information exchange harmonization, should gradually ease the manual burden of chart abstraction. AI-assisted abstraction and natural language processing are likely to mature into standard tools once paired with adequate human validation. Regional collaboration — shared registries, harmonized protocols, and pooled pharmacovigilance networks — could help smaller countries reach the scale that regulatory-grade evidence requires. And as national regulators such as EFDA build their own RWE review capacity, locally generated evidence is likely to carry more direct weight in approval and coverage decisions, not just as supporting context.

Bottom line: Regulatory-grade RWE comes down to a clear causal question, data that is fit for purpose, methods locked in ahead of time, real bias control, documented provenance, and transparent reporting. For Ethiopia and other LMICs, the real opportunity is turning the health data already being generated every day into credible local evidence, evidence that can make medicines safer, care better, and public health decisions sounder.

Glossary of Key Terms

Term	Definition
RWD (Real-World Data)	Data on patient health status or care collected outside conventional clinical trials — e.g., EHRs, claims, registries, pharmacy and lab records.
RWE (Real-World Evidence)	Clinical evidence about a medical product's use, benefits, or risks, derived from analysis of RWD.
Target trial emulation	A design discipline that specifies eligibility, treatment strategies, and analysis as if running a randomized trial, then emulates it using observational data.

Term	Definition
Estimand	A precise description of the treatment effect being estimated, defined by population, treatment, endpoint, intercurrent events, and summary measure (ICH E9(R1)).
External control arm	A comparator group built from existing data (e.g., historical or registry patients) rather than concurrent randomization, used to contextualize single-arm trial results.
Fit-for-purpose data	Data whose relevance and reliability are adequate to answer a specific research question, established through documented evaluation.
SAP (Statistical Analysis Plan)	A pre-specified document detailing planned statistical methods, finalized before analysis begins.
Pharmacovigilance (PV)	The science of detecting, assessing, and preventing adverse effects of medicines, often using post-approval RWD.
HTA (Health Technology Assessment)	A systematic evaluation of a medicine's clinical and economic value used to inform coverage and reimbursement decisions.

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Ten Medicinal-Chemistry Issues that Restrict Leads from Progressing to Useful Drugs

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Abstract

Medicinal chemistry rarely fails because a molecule cannot be made to bind its intended target. The harder problem is simultaneously achieving selectivity, exposure at the site of action, acceptable pharmacokinetic properties, safety, chemical stability, appropriate formulation and/or a practical economically feasible manufacturing method. The literature repeatedly shows that physicochemical properties, especially lipophilicity, polarity, solubility and molecular size are tightly coupled to absorption, distribution, metabolism, and excretion (ADME), selectivity and toxicological risk. The most useful aspect is framing a multi-parameter optimization under competing constraints.

Ten recurring issues that are development bottlenecks are listed in the table below. They are ordered broadly from core molecular properties through ADME, safety and manufacturability. A candidate can fail at more than one of these simultaneously, and some liabilities can be mitigated by formulation, prodrug design, dosing strategy or patient selection rather than by changing the parent structure.

In the appendix, example of adverse properties that hindered development of few potentially drug-resistant antibiotics is summarized to illustrate the title of this discussion.

List of Ten-bottlenecks That Restrict Progress from Leads to Drugs*

#	Description of Properties	Description of Issues	Effect on drug Development
1	Physicochemical imbalance	High lipophilicity, poor aqueous solubility, excessive polarity or crystal-lattice strength	Poor oral exposure, formulation difficulty, promiscuity or toxicity
2	Permeability & efflux	Insufficient membrane passage or active efflux	Low absorption or inadequate tissue/target-site exposure
3	Metabolic instability / clearance	Rapid CYP/UGT metabolism, renal clearance or metabolic soft spots	Short half-life, high dose, variable exposure
4	DDI and transporter liability	CYP inhibition/induction or transporter interactions	Drug-drug interactions, dosing restrictions, regulatory concern
5	Selectivity / off-target pharmacology	Potent binding to related or unrelated proteins	Narrow therapeutic window or mechanism-based toxicity
6	Target-site distribution	Poor penetration into required tissue, organelle or compartment	Adequate plasma PK but inadequate pharmacology where it matters
7	Chemical instability / reactive metabolites	Hydrolysis, oxidation, covalent reactivity or unstable metabolites	Loss of exposure, analytical complexity, safety risk
8	Organ-specific safety liabilities	hERG/QT, mitochondrial, phospholipidosis, DILI or other mechanistic liabilities	Preclinical/clinical safety termination or dose limitation
9	Therapeutic-index / dose feasibility	Efficacy requires exposures close to toxic or insoluble levels	No clinically workable dose or schedule
10	Synthetic & formulation developability	Complex synthesis, unstable solid form, poor crystallinity or difficult scale-up	Cost, supply, reproducibility or product-quality barriers

*CYP = Cytochrome P450; UGT = UDP-Glucuronosyltransferase; DDI = Drug-drug interaction; PK = Pharmacokinetics; hERG = *Ether-à-go*-Related Gene; DILI = Drug Induced Liver Injury

How these issues were identified

This review is based on published medicinal-chemistry reviews, drug-attribution analyses, regulatory guidance and contemporary property-prediction literature. The emphasis is on problems that are directly addressable—or at least strongly influenced—by molecular design. The evidence base includes analyses of physicochemical properties and attrition, guidance on QT and drug–drug interactions, and reviews of computational property prediction.

Drug-like-property analyses: Leeson & Springthorpe emphasized the relationship between increasing lipophilicity, selectivity and toxicological attrition.

Attrition analyses: industry-wide and multi-company analyses have linked compound properties and absorption, distribution, metabolism, excretion, and toxicity (ADMET) quality with progression.

Regulatory science: International Council for Harmonization of Technical Requirements for Human Use/ Food and Drug Administration (ICH/FDA) QT guidance and ICH M12 drug–interaction guidance illustrate how specific molecular liabilities become development constraints.

Modern property prediction: machine-learning reviews emphasize that simultaneous optimization of potency, solubility, permeability and ADMET remains a multi-objective problem with important data limitations.

1. Physicochemical imbalance: solubility, lipophilicity, polarity and solid-state behavior

Why it blocks development. Potency optimization often adds hydrophobic surface, aromatic rings and/or hydrogen-bonding capacity. Those changes can make a molecule bind more strongly while simultaneously reducing aqueous solubility, increasing nonspecific binding and making absorption or formulation harder.

Typical medicinal-chemistry manifestations. High cLogP/logD (calculated partition coefficient/distribution coefficient), low kinetic or thermodynamic solubility, high melting point, strong crystal packing, excessive H-bonding or high polar surface area. The apparent contradiction is that increasing polarity may improve solubility but can reduce permeability; reducing lipophilicity can reduce potency or tissue penetration.

Practical chemistry levers. Reduce unnecessary lipophilicity; replace flat aromatic systems with saturated or conformationally richer motifs; tune ionization; disrupt crystal packing when appropriate; use matched molecular-pair analysis; measure both kinetic and thermodynamic solubility; consider salt, co-crystal or prodrug strategies when structure optimization alone is insufficient.

Development consequence. Poor solubility can cap oral dose and exposure. Excess lipophilicity can increase promiscuity and safety risk. This is a central multi-parameter bottleneck rather than a single threshold problem.

Evidence / further reading. [Leeson & Springthorpe \(2007\)](#); [ACS: Designing Better Drug Candidates](#)

2. Permeability and active efflux

Why it blocks development. A compound may be potent in a biochemical assay but fail to reach the relevant biological compartment. Passive permeability is determined by size, lipophilicity, polarity, ionization and membrane interactions, while transporters can actively oppose uptake.

Typical medicinal-chemistry manifestations. Low apparent permeability, high efflux ratios, strong P-gp/BCRP substrate behavior, permeability that is highly pH-dependent, or poor intestinal uptake despite adequate solubility.

Practical chemistry levers. Balance lipophilicity and polarity rather than maximizing either; reduce exposed H-bond donors where justified; constrain conformational flexibility; optimize ionization; identify transporter liabilities early; use permeability and efflux assays in parallel with solubility.

Development consequence. The consequence can be low oral bioavailability or a large gap between plasma exposure and target-site exposure. Efflux is especially important for the intestine, liver, kidney and blood–brain barrier.

Evidence / further reading. [EMA drug-interaction guideline](#)

3. Metabolic instability and excessive clearance

Why it blocks development. Metabolism can erase otherwise excellent pharmacology by converting the parent compound too quickly. Attempts to block metabolism can create a cycle in which added lipophilicity or steric bulk introduces new liabilities elsewhere.

Typical medicinal-chemistry manifestations. Oxidation at benzylic/allylic positions, N- or O-dealkylation, aromatic hydroxylation, glucuronidation, renal clearance, or species differences that make human clearance difficult to predict.

Practical chemistry levers. Map metabolic soft spots early; use deuteration selectively where scientifically justified; replace labile groups; tune lipophilicity and ionization; exploit steric shielding or conformational restriction; test metabolites for activity and safety rather than focusing only on parent exposure.

Development consequence. High clearance can force frequent dosing or high doses, increasing peak/trough variability and safety burden. Over-correction can also produce very long half-life or metabolite accumulation.

Evidence / further reading. [ACS prodrug webinar—ADMET strategies](#)

4. CYP, UGT and transporter drug–drug interaction liability

Why it blocks development. A candidate can be pharmacologically effective yet unusable in common patient populations if it strongly inhibits or induces enzymes or transporters that control exposure to co-medications.

Typical medicinal-chemistry manifestations. Time-dependent CYP inhibition, mechanism-based inhibition, CYP induction, transporter inhibition, transporter substrate behavior, or strong dependence on a single clearance pathway.

Practical chemistry levers. Avoid unnecessary planar hydrophobic motifs associated with promiscuous binding; reduce affinity for major CYPs where possible; balance clearance pathways; test major CYPs and transporters early; use mechanistic static models and physiologically based pharmacokinetic modelling (PBPK) when justified.

Development consequence. DDIs can require dose restrictions, contraindications, monitoring or a narrower label. ICH M12 explicitly addresses CYP, UGT and transporter-based DDI evaluation and predictive modeling.

Evidence / further reading. [ICH M12 drug-interaction guideline](#)

5. Insufficient selectivity and off-target pharmacology

Why it blocks development. Increasing affinity at the desired target does not guarantee selectivity. Closely related proteins, ion channels, G-protein-coupled receptors (GPCRs), nuclear receptors and other high-capacity binding sites can recognize the same physicochemical features.

Typical medicinal-chemistry manifestations. Activity in broad safety panels, receptor/channel binding, hERG activity, CYP inhibition, phospholipidosis, or phenotypic effects inconsistent with the intended mechanism.

Practical chemistry levers. Improve ligand efficiency rather than simply increasing hydrophobic contacts; exploit less-conserved binding pockets; use structure-based design and selectivity panels early; distinguish target-mediated pharmacology from nonspecific membrane or protein binding.

Development consequence. Off-target activity can lower the therapeutic index and create clinical monitoring or exclusion criteria. Lipophilicity is a recurring risk amplifier because it can increase nonspecific interactions.

Evidence / further reading. [Leeson & Springthorpe \(2007\)](#)

6. Inadequate target-site and tissue distribution

Why it blocks development. Plasma exposure is not equivalent to exposure at the molecular target. The required site may be behind the blood–brain barrier, inside cells, within an organelle, in a poorly perfused tumor, or in a protected inflammatory compartment.

Typical medicinal-chemistry manifestations. Good systemic PK but weak pharmacodynamic biomarker response; poor free-drug exposure at the target; excessive sequestration in fat, lysosomes or plasma proteins; transporter-mediated exclusion from the target tissue.

Practical chemistry levers. Design for the target compartment explicitly; measure free rather than total exposure; use tissue-to-plasma ratios cautiously; integrate permeability, transporter and binding data; consider targeted delivery or prodrug approaches where appropriate.

Development consequence. This issue can produce apparent 'efficacy failures' even when target engagement is demonstrated in vitro. It is particularly important for CNS and intracellular targets.

Evidence / further reading. [ACS prodrug webinar—site-selective and exposure strategies](#)

7. Chemical instability and reactive metabolites

Why it blocks development. A candidate may degrade during formulation, storage, assay or dosing, or form electrophilic/reactive metabolites after metabolism. Such liabilities can be difficult to separate from desired pharmacology.

Typical medicinal-chemistry manifestations. Hydrolysis, oxidation, photolysis, tautomerization, covalent reactivity, unstable intermediates, glutathione adducts or metabolites that bind proteins or DNA.

Practical chemistry levers. Remove or attenuate chemically labile motifs; identify degradation products and major metabolites early; use stability-indicating analytics; assess covalent reactivity and reactive-metabolite formation; retain structural alerts as hypotheses to test, not automatic rejection criteria.

Development consequence. Chemical instability can compromise dose consistency and interpretation of pharmacology. Reactive metabolites can create idiosyncratic or organ-specific safety concerns that are hard to predict perfectly from structure alone.

Evidence / further reading. [ACS: Reactive metabolites in drug discovery](#)

8. Mechanism-based organ toxicity and electrophysiology liabilities

Why it blocks development. Some molecular mechanisms produce safety signals at exposures that overlap the desired pharmacology. Medicinal chemistry can reduce—but not eliminate—these risks.

Typical medicinal-chemistry manifestations. *Ether-à-go-Related Gene*/Rapidly activating delayed-rectifying K⁺ current (hERG/Ikr) inhibition and QT prolongation, mitochondrial toxicity, phospholipidosis, bile-salt transporter effects, liver injury mechanisms, or tissue-specific accumulation.

Practical chemistry levers. Run focused safety screens early; lower hERG affinity without sacrificing target potency; reduce amphiphilic character where phospholipidosis is a concern; avoid problematic motifs when alternatives exist; characterize exposure margins using free concentrations.

Development consequence. A safety signal can terminate a program or impose a dose ceiling. FDA/ICH E14-S7B guidance specifically addresses the nonclinical and clinical evaluation of QT/QTc and proarrhythmic potential.

Evidence / further reading. [FDA/ICH E14-S7B QT guidance](#)

9. Therapeutic-index and dose-feasibility failure

Why it blocks development. The clinically relevant question is not whether a molecule works, but whether it can produce sufficient target engagement at a tolerable exposure. Medicinal chemistry often faces a three-way conflict among potency, exposure and safety.

Typical medicinal-chemistry manifestations. Potency is excellent but oral dose would exceed solubility; efficacy requires concentrations near toxicity thresholds; efficacy is lost before exposure reaches the target; active metabolites complicate exposure margins.

Practical chemistry levers. Optimize free-drug potency and exposure together; use translational pharmacokinetic/pharmacodynamic (PK/PD) models; prioritize unbound exposure; design around exposure margins early; avoid unnecessary molecular inflation that raises dose and clearance.

Development consequence. A compound with strong in vitro activity can still be a non-drug if the projected human dose is impractical or the therapeutic window is too narrow.

Evidence / further reading. [Machine-learning review of multi-property optimization](#); [ACS: Designing Better Drug Candidates](#)

10. Synthetic, solid-form and formulation developability

Why it blocks development. A molecule that meets biological criteria may still be a poor drug candidate if it requires an impractical synthesis, is difficult to purify, has unstable stereochemistry, forms an undesirable solid form, or cannot be formulated at the required dose.

Typical medicinal-chemistry manifestations. Many synthetic steps, chromatography-dependent purification, unstable intermediates, difficult regio-/stereochemical control, polymorphism, hygroscopicity, poor crystallinity, oiling-out, low yield or a formulation that cannot deliver the projected dose.

Practical chemistry levers. Design with route and scale in mind; reduce step count and protecting-group burden; identify robust crystalline forms; screen salts/co-crystals; characterize polymorphs; assess chemical and physical stability; use scalable catalytic or flow-compatible chemistry where appropriate.

Development consequence. These liabilities can delay Investigational New Drug (IND)-enabling work, increase Chemistry, Manufacturing and Controls (CMC) risk, create batch-to-batch variability or make a promising molecule commercially impractical.

Evidence / further reading. [Preformulation and candidate-selection overview](#)

11. Optimization strategy

Optimize properties concurrently. Potency-first optimization followed by a late ADMET 'cleanup' phase is often inefficient because the properties are mechanistically coupled.

Treat lipophilicity as a system variable. It influences solubility, nonspecific binding, permeability, CYP interactions, tissue distribution and toxicity; reducing it can improve several dimensions at once, but only if target-site exposure is preserved.

Measure the right quantity. Total plasma concentration can be misleading when protein binding is high; unbound concentration and target-site exposure often provide a more useful bridge from in vitro potency to in vivo pharmacology.

Use orthogonal assays. A single permeability, solubility or toxicity assay rarely explains the full mechanism. Confirm liabilities with complementary systems and, where possible, human-relevant models.

Design around mechanisms, not labels. 'Rule of five' and structural-alert heuristics are useful filters, not laws. Exceptions are common, particularly for transporters, central nervous system (CNS) drugs,

macrocycles, natural-product-like scaffolds and other beyond-Rule-of-5 modalities.

Build manufacturability into lead optimization. Synthetic accessibility, solid form and formulation should become increasingly explicit design objectives as a program approaches candidate selection.

12. Examples of Adverse properties that hinder development of antibiotics designed to resist resistance

This appendix adds a focused discussion of research antibiotics and antibiotic strategies that are notable because they either attack unusually conserved/immutable bacterial structures, use mechanisms that bypass established resistance, or seek to resensitize resistant organisms. However, it should be noted that no antibiotic can currently be described as guaranteed to 'never develop resistance'.

1. Teixobactin

Development status. Teixobactin was reported in *Nature* in 2015 after discovery from an uncultured soil bacterium. It is active against important Gram-positive pathogens and showed *in-vivo* activity in early animal studies. The original work reported no detectable resistance under the experimental conditions used.

Why resistance may be unusually difficult to develop. Teixobactin binds lipid II and lipid III, cell-envelope precursors rather than a conventional mutable protein target. The conceptual advantage is that the bacterium has fewer sequence-level options for escaping interaction. It also damages the membrane through supramolecular assembly, creating more than one pharmacological pressure.

Key barriers to useful drug development. Teixobactin is not a ready-made drug. Its peptide nature creates challenges in stability, exposure, delivery and manufacturing. Its activity is strongest against Gram-positive organisms, so it does not directly solve the entire Gram-negative antimicrobial drug resistance (AMR) problem. Translation from mouse models to human pharmacology and practical clinical formulation remain major hurdles.

Key references. [Ling et al., Nature 2015](#); [ASM Microcosm review](#)

2. Darobactin

Development status. Darobactin was discovered from bacteria associated with entomopathogenic nematodes and reported as a new antibiotic selectively killing Gram-negative pathogens. It acts on BamA, an essential outer-membrane protein.

Why resistance may be unusually difficult to develop. Darobactin recognizes an extracellular region of BamA through an unusual peptide-like binding mode. Published work found resistant mutants at very low frequency and showed that mutations clustered in the binding region, indicating that escape can require substantial structural disruption.

Key barriers to useful drug development. Darobactin is a ribosomally synthesized and post-translationally modified peptide. Development therefore faces the classic challenges of peptide stability, pharmacokinetics, production and formulation. Its narrow biological spectrum and the possibility of target mutations also mean that 'low resistance' should not be equated with resistance-proof.

Key references. [Imai et al., Nature 2019](#); [Kaur et al., Nature 2021](#)

3. Clovibactin

Development status. Clovibactin was reported in Cell in 2023 after isolation from an uncultured bacterium. It is active against drug-resistant Gram-positive pathogens and demonstrated in-vivo activity in animal models.

Why resistance may be unusually difficult to develop. Clovibactin binds the pyrophosphate component of multiple essential cell-wall precursors, including lipid II and wall-teichoic-acid precursors. The target moiety is highly conserved, and the compound forms supramolecular structures on bacterial membranes. The reported spontaneous resistance frequency was extremely low.

Key barriers to useful drug development. Clovibactin is still an experimental peptide antibiotic. Its physicochemical properties, formulation, stability, exposure, scalable synthesis/production and clinical safety need to be solved. Its current spectrum also limits it as a universal AMR solution.

Key references. [Shukla et al., Cell 2023](#); [Nature Reviews / perspective on innovative antibiotic discovery](#)

4. Zosurabalpin (RG6006)

Development status. Zosurabalpin is a Roche antibiotic candidate directed at carbapenem-resistant *Acinetobacter baumannii*. Roche reported that the program had advanced toward Phase III, with Phase III initiation planned for 2026 in company pipeline materials.

Why resistance may be unusually difficult to develop. The molecule targets the LptB2FGC system responsible for lipopolysaccharide transport. This is a novel antibacterial mechanism, and the compound recognizes a composite site involving the transporter and its LPS substrate. Such a mechanism bypasses many resistance mechanisms that act on established antibiotic targets.

Key barriers to useful drug development. The key limitation is that zosurabalpin is specialized rather than broad-spectrum: its development is centered on carbapenem-resistant *A. baumannii*. Resistance to a new antibiotic remains biologically possible, so surveillance will be required. Clinical efficacy, resistance emergence, IV dosing, optimal use with other agents and stewardship all remain important development questions.

Key references. [Zampaloni et al., Nature 2024](#); [Pahil et al., Nature 2024](#); [Roche pipeline material](#)

5. Cresomycin

Development status. Cresomycin is a synthetic macrobicyclic antibiotic reported in Science in 2024. It was designed from the oxepanoprolinamide scaffold to bind the bacterial ribosome in a preorganized conformation.

Why resistance may be unusually difficult to develop. The preorganized structure improves ribosomal binding and allows activity against bacteria carrying methyltransferase-based resistance mechanisms that defeat several older ribosome-targeting antibiotics. This is an example of medicinal chemistry being used specifically to design around an established resistance mechanism.

Key barriers to useful drug development. The compound remains preclinical. Major questions include human exposure, toxicity, formulation, oral bioavailability, resistance selection in clinical populations and manufacturing. Strong biochemical or animal data do not yet establish a clinically useful therapeutic window.

Key references. [Cresomycin / Science report](#); [CARB-X development announcement](#)

6. Lolamicin

Development status. Lolamicin was reported as a selective Gram-negative antibiotic targeting the LolCDE lipoprotein transporter. Its significance is partly conceptual: it was designed to exploit a bacterial-envelope pathway while attempting to spare the mammalian microbiome.

Why resistance may be unusually difficult to develop. The target is distinct from most established antibiotic classes, and the work demonstrated activity against multiple Gram-negative pathogens, including resistant strains. A novel target can reduce cross-resistance, although it does not eliminate the possibility of de-novo resistance.

Key barriers to useful drug development. Development remains early. Key barriers include establishing human PK/PD, therapeutic window, safety, spectrum, tissue penetration, resistance frequency and resistance pathways. The promising microbiome-selectivity observation also needs confirmation in clinical populations.

Key references. [Lolamicin research report](#)

7. SCH-79797

Development status. SCH-79797 is a research antibiotic with a dual mechanism: inhibition of folate metabolism together with disruption of bacterial membrane integrity. It has shown activity against multiple clinically important pathogens in preclinical work.

Why resistance may be unusually difficult to develop. Dual mechanism can make simple single-target resistance less effective because the bacterium would need to tolerate or bypass two distinct pressures. This is a useful design principle for future antibiotic discovery rather than evidence that SCH-79797 itself is resistance-proof.

Key barriers to useful drug development. The main limitation is that it has not progressed as a conventional clinical antibiotic. Mechanistic complexity, selectivity, mammalian toxicity, PK and formulation must be solved before a human therapeutic program could be justified.

Key references. [Innovative perspectives on small-molecule antibiotic discovery](#)

8. Amit Singh's TB resistance-resensitization strategy

Development status. Dr Amit Singh's IISc research is important but is better described as a resistance-reversal strategy than as a single 'new antibiotic.' His group investigated how *M. tuberculosis* changes redox/oxidative-stress physiology during isoniazid resistance and identified compounds such as ebselen and vancomycin that, in combination with isoniazid, increased killing of resistant strains in experimental systems. His group also investigated chloroquine plus

isoniazid to alter the intracellular environment and shorten therapy in animal models.

Why resistance may be unusually difficult to develop. The central idea is to target an adaptive vulnerability created or exposed by resistance rather than continually replacing one antibiotic with another. This can potentially make an existing antibiotic effective again and may reduce the selective pressure associated with simply escalating exposure to a single agent.

Key barriers to useful drug development. The major barriers are translational. Combination regimens require compatible human PK, safety, dosing and interaction profiles. TB drugs must reach intracellular and lesion compartments, treatment must be long enough to prevent relapse, and resistance to the combination itself must be monitored. The published work is not evidence that resistance can never develop.

Key references. [DBT India Alliance report on Amit Singh's work](#); [IISc Connect feature on Singh's TB work](#); [Amit Singh laboratory / IISc context](#)

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THE POTENTIALITIES OF TRADITIONAL HEALERS AND INDIGENOUS DRUGS IN HEALTHCARE IN AFRICA

13TH ANNUAL MEETING OF THE ASSOCIATION OF MEDICAL SCHOOLS IN AFRICA; APRIL 23 - 27, 1979
ADDIS ABABA UNIVERSITY FACULTY OF MEDICINE

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(Continued from Volume 6, Number 1, 2006)

TABLE I : PLANTS USED AS ANTHELMINTHS

Vernacular name	Scientific name	Part of plant
1. AZAMR	Bersama abyssinica	Bark of root
2. AMWASHERO	Caralluma Sprengeri	Root
3. ANTERFA	Euphorbia longecornuta	Root
	E.petitiana	
	E.longepetiolata	
4. ENQO00	Embelia Schimperii	Seed
5. ET	Malva Parviflora	Root
6. ROMAN	Punica Granatum	Fruit & bark of root
7. SHNA	Albizia Anthelmintica	Bark of Stem
8. QECHEMO	Myrsine Africana	Seed
9. QURQURA	Zizyphus Spina-Christi	Seed
10. BSANNA	Croton Macrostachys	Young growth & Bark of Stem
11. DEDHO	Euclea Schimperii	Root
	E.Divinorum	
12. DUBBA	Cucurbita Pepo	Seed
13. GZAWA	Withania Somnifera	Bark of root
14. GUAZZA	Dioscorea SP.	Root
15. TELAZZA	Gomphocarpus Fruticosus	Root
16. TELENJ	Achyranthes Aspera	Root
17. FATUQA	Rhus Matalensis	Seed

18. WEGGERT	Silene Macroselen	Root
19. AMFAR	Buddleja Polystachya	Bark of root
20. YEOUOLLA ENDEHWAHWLLA	Kalenchoe Lanceolata	Bark of root
21. TENBELEL	Jasminum Floribundum	Root
22. TINJUT	Otostegia Integrifolia	Root
23. ITSE MENAHIE	Securidaca Longinedunculata	Bark of root
24. ITSE SABEQ	Zehneria Scabra	Root
25. ADENDASH	Adenia Venenata	Root
26. ADES	Myrtus Communis	Root
27. LIMMITCH	Clausena Anisata	Root
28. SHDEDO	Pennisetum Schimperii	Root
	P. Glabrum	
29. AKERMA	Eleusine Jaegeri	Root
	E.Floccifolia	
30. ZGBA	Podocarpus Gracilior	Root
31. GOMMEN ZER	Brassica Carinata	Seed
32. GRANTA	Cyperus Lonqus	Root
	C. Rigidifolius & other similar species	
33. ATTUCH	Verbena Oficinalis	Root
34. KTKTTA	Dodonea Viscosa	Root
35. MENMENNO	Rumex Abyssinicus	Root
36. YEWSKA DEMBLAL	Salvia Schimperii	Seed
37. YEFYEL ZOAOBIE	Ocimum Suave	leaf & seed
38. WANZA	Cordia Africana	Seed & Bark of root
39. BETRE MUSIE	Grewia Spp.	Seed
40. QECHEMO	Myrsine Africana	Seed
41. BEDDENO	Balanites Aegyptiaca	Seed & Bark of root
42. QULQUAL	Euphorbia Abyssinica	Flower E.Candelabrum
43. KOSHM	Dovyalis Abyssinica	Fruit
44. TULT	Rumex Beaquertii	Root

45. ANANAS	Annanas Commosus	Juice
46. YELAM TUT	Eulophia Schimperiana	Root
47. FRQA	Piliostigma Thoningi	Root
48. ITSE YESUS	Stefania Abyssinica	Root
49. IRGUFT	Erythrina Abyssinica	Bark of root
50. QENQELCHIA	Aeschynomene Elaphroxylon	Root Bark of root
51. QOQOBA	Maytenus Senegalensis	Bark of root
52. TEDDO	Rhamnus Staddo	Young growth
53. YEMDR IMBUAY	Cucumis Ficifolius	Root
	C.Prophetarum	
54. MSRCH	Clerodendron Myrtcoides	Root
55. KOILO	Echinops Steudneri	Inflorescence
	E.Amplexicaulis	
	E.Ellenbeckii	
56. ERET	Aloe spD.	Sap

TABLE 2. List of Plants used in the Treatment of various kinds of NEQERSA

Note that NEQERSA does not fully correspond with cancer. The delimitation is, however, not yet well studied though we know, for example, that tuberculosis is also NEQERSA

A. KEISI NEQERSA

VERNACULAR	NAME SCIENTIFIC	NAME PART USED
1. ITSE MANAHIE	Securidaca longipedunculata	Root
2. ITSE ANBESSA	Asplenium Aethiopicum	Rhizoids
3. ITSE MEFRH	Cineraria Abyssinica	Whole plant & other spp.
4. JEBBERA	Lobelia Rhynchoptallum	Root
5. ALEBLABIT	Traquia Pungens	Root
6. INBUATCHO	Rumex Nervosus	Root
7. WENDIE QATCHA	Sanseveria sp.	Root
8. AMERA	Plumbago Zeylanicum	Root

9. MEQMEQO	Rumex Abyssinicus	Root
10. MEREZ	Acokanthera Schimperii	Root
11. DEDEHO	Euclea Schimperii	Root
12. QULQUAL	Euphorbia Abyssinica	Latex
	E.Candelabrum	
	E.Obovalifolia	
13. QNBO	Calotropis Procera	Latex
14. ZINJERO ALENGA	Sarcostema spp.	Latex
15. MOIDER	Periploca Linearifolia	Latex
16. QNTCHB	Euphorbia Tirucall	Latex
17. ANTERFA	Euphorbia Longecornuta	Latex
	E.Petitiana	
	E.Longipetiolata	
18. WEFTETCH	Momordica Foetida	Leaf
19. KTKTTA	Dodonea Viscosa	Leaf
20. IMBWATCHO	Rumex Meruosus	Leaf
21. TSEDDO	Rhamnus Staddo	Leaf
22. YEHIFATKITEL	Conyza Steudeli	Leaf
23. GINDOSH	Cyphostemma Sp.	Root
24. SREBZU	Thalictrum Rhynchocharpum	Root
25. DOG	Ferula Communis	Root
26. TIT	Gossypium Barbadense	Seed
	G.Herbaceum	
27. SUF	Carthamus Tinctorius	Seed
28. NUOG	Guizotia Abyssinica	Seed
29. SELIT	Sesamum Indicum	Seed

B. MNSHRO NEQERSA

Vernacular name	Scientific name	Part of plant
1. MEQMEQO	Rumex Abyssinicus	Root & young growth

2. MSSRCH	Clerodendron Myricoides	Root & Flower
3. MEREZ	Acokanthera Schimperii	Leaf, fruit young growth & flower
4. SREBZU	Thalictrum Rhynchocarpum	Leaf & fruit
5. SENSEL	Adhatoda Schimperiana	Root & young growth
6. SENAFTCH	Brassica Nigra	Seed
7. TIQUR SNDIE	Triticum Aestivum	Seed
8. RAS KMMR	Leonotis Velutina	Root & leaf
	L. Raineriana	
9. ROMAN	Punica Granatum	Bark & leaf
10. SHUNSHUNNA	Debregeasia Bicolor	Bark
11. SHNET	Myrica Salicifolia	Bark & root
12. YEQOLLA SHOLA	Ficus spp.	Fruit & bark of root
13. QULQUAL	Euphorbia Abyssinica	Latex & root
	E.Candelabrum	
	E. Obovalifolia	
14. OENTAFFA	Pterolobium Stellatum	Root & leaf
15. WENDIE QATCHA	Sansevieria sp.	Root
16. QETETNNA	Verbascum Sinaiticum	Root
17. QESTENCH	Asparagus Africanus	Root
18. QERET	Ossyris Abyssinica	Bark
19. QNBO	Calotropis Procera	Latex, leaf & root
20. BELES	Ficus Palmata	Latex, root, leaf & fruit
21. BAQELA	Vicia Faba	Root, Flower
22. BUNNA	Coffea Arabica	Leaf, seed
23. BERBERE	Capsicum Frutescens	Leaf & fruit
24. BRBRRA	Milletia Ferruginea	Seed
25. TALO	Rhus Retinorrhoea	Root & seed
26. TEQETSLLA various hosts	Tapinanthus spp. & others in Loranthaceae, Excluding VISCUM, & growing on	Whole plant
27. MDR INBUAY	Cucumis Prophetarum	Root

	C. Ficiflius	
28. GEBER IMBUAY	Solanum Margintum	Root & fruit
29. MEKAN ENDOD	Phytolacca Dodecandra	Root
30. INSLAL	Foeniculum Vulgare	Root & Young growth
31. AZAMR	Bersama Abyssinica	Root
32. ITSE SIOL	Ranunculus multifidus	Leaf
33. ANTERFA	Euphorbia longecornuta	Root & leaf
	E. Petitioner	
	E. Longipetiolata	
34. ATKUAR	Nuxia conjeſta	Root & young growth
35. ITSE MENAHIE	Securidaca longepedunculata	Root
36. AMEDMADDO	Chenopodium ambrosoides	Root & leaf
37. AGAM	Carissa edulis	Root, leaf & flower
38. ASTA	Erica arborea	Root & flower
39. AMERA	Plumbago zeylanicum	Root, leaf & flower
40. ITSE ANBESA	Asplenium aethiopicum	Leaf & Rhizoids
41. SETIE RIET	Aloe spp	Root
42. WEFIE GOMMEN	Senecio cf.vulgaris	Root & leaf
43. INBATCHO	Rumex nervosus	Root
44. ITSE SABEQ	Zehneria scabra	Root
45. ITSE MEFRH	Cineraria abyssinica	Root & other spp.
46. ITS EYEHAYU	Habenaria sp.	Tuber
47. ANFAR	Buddleja polystachya	Leaf
48. ADENDASH	Adenia venenata	Root
49. ITSE YESUS	Stephania abyssinica	Root
50. AKRMA	Eleusine jaegeri & similar spp	Root.
51. AMJJA	Hypericum revolutum	Root & seed
52. AMOTCH	Sauromatum sp.	Root & fruit
	Amorphophallus sp.	
53. HINNA	Lawsonia inermis	Leaf

54. LOL	<i>Ekebergia capensis</i>	Leaf & root
55. LT	<i>Malva parviflora</i>	Leaf & root
56. LMMTCH	<i>Clausena anisata</i>	Root, leaf
57. YETIJA TCHENGER	<i>Hibiscus micranthus</i>	Root & flower
58. TINBAHO	<i>Nicotiana tabacum</i>	Root & flower
59. NETCHLLO	<i>Senra incana</i>	Root
60. TENA ADDAM	<i>Ruta chalpensis</i>	Fruit
61. NATCHA	<i>Kosteletzkyia adoensis</i> <i>K. longifolia</i>	Young growth & flower
62. KOSO	<i>Hagenia abyssinica</i>	Root, leaf, seed, flower
63. WAGINOS	<i>Brucea antidysenterica</i>	Seed
64. WANZA	<i>Cordia africana</i>	Root & fruit
65. WEIN	<i>Vitis vinifera</i>	Root & bark
66. WEIRA	<i>Olea africana</i>	Root&bark
67. WLKFFA	<i>Dombeya spp.</i>	Root & bark
68. ZNJBL	<i>Zingiber officinala</i>	Corms
69. YEJIB MRKUZ	<i>Heteromorpha trifoliata</i>	Root
70. YEYIFATOTEL	<i>Conyza steudelii</i>	Root
71. DOG	<i>Foeniculum vulgare</i>	Root & Inflorescenc
72. DGTTA	<i>Calpurnia aurea</i>	Root, leaf, flower, seed
73. DEDHO	<i>Euclea schimperi</i>	Root, flower,bark, flower
74. DNBLAL	<i>Coriandrum sativum</i>	Root, seed
75. DOQMA	<i>Syzygium guineense</i>	Bark
76. DUADUATIE	<i>Colutea istria</i>	Seed
77. JEBBERA	<i>Lobelia rhynchopetallum</i>	Root, flower
78. GRAWWA	<i>Vernonia amygdalina</i>	Root, young growth
79. ASHKOKKO GOMMEN	<i>Senecio gigas</i>	Root
80. GUAYYA	<i>Lathyrus sativa</i>	Seed
81. GZAWA	<i>Withania somnifera</i>	Leaf
82. GESHO	<i>Rhamnus prinoides</i>	Fruit

83. GETEM	<i>Schefflera abyssinica</i>	Bark of root
84. GUERTEB	<i>Plantago lanceolata</i>	Root & leaf
85. YEBEREHA GULO	Wild forms of <i>ricinus communis</i>	Seed
86. TID	<i>Juniperus procera</i>	Fruit
87. TOSSGN	<i>Thymus serrulatus</i>	Leaf
88. TIT	<i>Gossypium barbadense</i>	Seed, root
	<i>G. herbaceum</i>	
89. TOBBYA	<i>Calotropis procera</i>	Root, flower
90. TENBELEL	<i>Jasminum floribundum</i>	flower
91. TELENJ	<i>Achyranthes aspera</i>	Root & leaf

C. Kemenz Neqersa

Vernacular name	Scientific name	Part of plant
1. Indehwahulla	<i>Kalenchoe</i> spp.	Leaf & root
2. Amjja	<i>Hypericum revolutum</i>	Roots & seed
	<i>H. Quartinianum</i>	
3. Atuch	<i>Verbena officinalis</i>	Root
4. Iriet	<i>Aloe</i> spp. Young	growth & root
5. Inbuatcho	<i>Rumex nervosus</i>	Young growth
6. Inguechit	<i>Stephania abyssinica</i>	Root
7. Geber Imbuay	<i>Solanum marginatum</i>	Root
8. Haregres	<i>Zehneria scabra</i>	Root & leaf
9. Merez	<i>Acokanthera schimperi</i>	Root, leaf & fruit
10. Mssrch	<i>Clerodendron myricoides</i>	Young growth & flower
11. Shunshunna	<i>Debregeasia bicolor</i>	Root & bark
12. Talo	<i>Rhus retinerrhoea</i>	Root
13. Netchlla	<i>Senra incana</i>	Root
14. Getem	<i>Schefflera abyssinica</i>	Bark
15. Dgndg	<i>Atriplex farinosa</i>	Root

16. Tid	<i>Juniperus procera</i>	Root & Seed
17. Tfie	<i>Olinia aequipetala</i>	Root & seed
18. Telenj	<i>Achyranthes aspera</i>	Root & young growth
19. Fielefej	<i>Clutia richardiana</i>	Root & leaf
	<i>C. Robusta</i>	

D. Yewist Neqersa

Vernacular name	Scientific name	Part of plant
1. Mehan indod	<i>Phytolacca dodecandra</i>	Root
2. Agam	<i>Carisa edulis</i>	Root
3. Indehwahulla	<i>Kalenchoe</i> spp	Root & leaf
4. Amera	<i>Plumbago zeylanicum</i>	Root
5. Anfar	<i>Buddleja polystachya</i>	Root
6. Inslal	<i>Foeniculum vulgare</i>	Root
7. Aleblabit	<i>Tragia pungens</i>	Root
8. Itse yodit	<i>Polygala</i> sp.	Root
9. Itse sabeq	<i>Zehneria scabra</i>	Root
10. Inzerezey	<i>Gladiolus pssittacinus</i>	Root & flower Bark
11. Inkoy.	<i>Ximenia americana</i>	Bark
12. Inbuatcho	<i>Rumex nervosus</i>	Leaf & root
13. Amedmaddo	<i>Chenopodium Ambrosoides</i>	Root & leaf
	<i>C. murale</i>	
14. Ashkt	<i>Galium spurium</i> & other spp.	Leaf
15. Yelomi wha	<i>Satureja</i> sp.	Leaf
16. Qestencha	<i>Asparagus africanus</i>	Root
17. Muatsh	<i>Duranta repens</i>	Root
18. Semeg	<i>Cardiospermum halicacabum</i>	Root
19. Roman	<i>Punica granatum</i>	Root
20. Shola	<i>Ficus</i> spp.	Bark

21. Shenbeqo	Arundo donax	Root
	Phragmites communis	
22. Qetetnna	Verbascum sinaiticum	Root
23. Qe ret	Ossyris abyssinica	Root & Bark
24. Qnbo	Calotropis procera	Root
25. Sete bsanna	Croton macrostachys	Root
26. Netchlo	Senra incana	Root & leaf
27. Natcha	Kosteletzkyia adoensis	Leaf & flower
	K. longifolia	
28. Kurncht	Oxygonum atriplicifolium	Root & leaf
29. Ktktta	Dodonea viscosa	Root & leaf
30. Wanza	Cordia africana	Root & leaf
31. Yewha anqur	Commelina spp	Leaf
32. Gortebb	Plantago Lanceolata	Root
33. Tobbia	Calotropis procera	Flower
34. Tenbelel	Jasminum floribundum	Leaf
35. Tchiqugn	Artemisia afra	Leaf
36. Fielefej	Clutia richardiana	Root, leaf, seed other spp.

E. Sanba Neqersa (Neqersa of the lungs)

Vernacular name	Scientific name	Part of plant
1. Inslal	Foeniculum vulgare	Root
2. Amera	Plumbago zeylanicum	Root
3. Inzerezei	Gladiolus pssittacinus	Root & leaf
4. Itse Menahe	Securidaca longipedunculata	Root
5. Asta	Erica arborea	Root & flower
6. Amotch	Sauromatum sp.	Tuber Amorphophallus spp.

7. Adendash	<i>Adenia venenata</i>	Root & bark
8. Iret	<i>Aloe sp.</i>	Flower & succulent part
9. Aguashr	<i>Plectranthus edulis</i>	Root & flower
10. Agam	<i>Carisa edulis</i>	Flower
11. Inbwatcho	<i>Rumex nervosus</i>	Young growth
12. Injorre	<i>Rubus apetalus</i>	Leaf
	<i>R. Volkensii</i>	
13. Argefgafo	<i>Protea gaugedi</i>	Root & leaf
14. Azamr	<i>Bersame abyssinica</i>	Root & young growth
15. Netch shinkurt	<i>Alium cepa</i>	Bulb
16. Tult	<i>Rumex beaqueritii</i>	Root & Leaf
17. Talo	<i>Rhus retinnorhoea</i>	Root
18. Bsanna	<i>Croton macrostachys</i>	Root & seed
19. Bunna	<i>Coffea arabica</i>	Leaf & seed
20. Berbere	<i>Capsicum frutescens</i>	Fruit
21. Qbie zelzl	<i>Romulaea campanuloides</i>	Leaf
22. Qulqual	<i>Euphorbia candelabrum</i>	Flower
	<i>E. abyssinica,</i>	
	<i>E. Obovalifolia</i>	
23. Merez	<i>Acokanthera schimperii</i>	Flower
24. Lenquata	<i>Grewia ferruginea</i>	Flower
25. Meqmebqo	<i>Rumex abyssinicus</i>	Root
26. Qega	<i>Rosa abyssinica</i>	Flower
27. Kok	<i>Prunus persica</i>	Root, flower
28. Kosso	<i>Hagenia abyssinica</i>	Root, seed & leaf
29. Wanza	<i>Cordia africana</i>	Root Fruit & leaf
30. Weira	<i>Olea africano</i>	Fruit & root
31. Zgba	<i>Podocarpus gracilior</i>	Root & gum
32. Znjbl	<i>Zingiber officinale</i>	Root
33. Dog	<i>Ferula communis</i>	Root & leaf

34. Dedho	Eucclea schimperi, E. Divinorum	Root & leaf
35. Jebbera	Lobelia rhynchopetallum	Root & flower
36. Tossn	Thymus serrulatus	Leaf
37. Telenj	Achyranthes aspera	Root
38. Tena adam	Ruta chalpensis	Fruit
39. Tchiret	Sansevieria sp.	Root
40. Tfrzila	Ceropegia affinis	Tuber
41. Werq bemeda	Ccecinia sp. (?)	Root
42. Yemdr imbuay	Cucumis prophetarum C. Ficifolius	Root

***Editors' Note: Tables 1, 2 and 3 were mentioned but left out in the article that was published in EPPAD Bulletin, Vol 6, No 1 (March 2006 issue). We were able to retrieve Tables 1 and 2 (above), but despite our efforts we could not locate Table 3. The original version was a rarely available mimeographed paper.**

